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Human Computer Interaction Labs 4th Year **2026/2027**

Lab 2 Semester: S7

Introduction to Human Computer Interaction: Design Rules

Design rules are universal principles that guide the creation of intuitive, efficient, and user-friendly interfaces. They are not tied to a specific platform or technology but emerge from psychology, design theory, and usability basics. These rules like *consistency, visibility, feedback, affordance, and constraints* help designers create systems (*Software and Hardware*) that users can understand and interact with naturally. By applying design rules, we ensure that a systems (*Software and Hardware*) not only looks good but works well in supporting human needs and tasks.

Objectives:

- Define and explain core *design rules*.
- Identify how these rules *appear or are broken*, in both *digital and physical interfaces*.
- Apply *design rules* to propose improvements in existing designs.
- Understand how ignoring rules creates *frustration* (*Learn by Critique + Correction*).
- **Work in groups of 3 max to generate more reliable results and more accurate findings related to *UI* and *UX*.**

Lab Work

1. **Useful Notions: Warming-Up Thoughts [10mins Disc]**

   It is important to note that design rules represent practical plan principles that guide the creation of usable systems. These rules are more concrete and focused on everyday interactions. They emphasize how users perceive, understand, and respond to a system. The table below summarizes four key rules: *Consistency, Visibility, Feedback, and Affordance & Constraints.*

Rule	Definition	Bad Example (Real Case)	Good Example (Real Case)
Consistency	Different layouts, icons, or terminology confuse users	*Instagram vs. TikTok navigation buttons* → Both apps are used for short-form videos, but their "Like," "Share," and "Comment" icons are placed differently and sometimes use different visuals. A new user switching between them may instinctively tap the wrong area, leading to confusion. This inconsistency increases cognitive load.	*Microsoft Office Ribbon (2007+)* → Before 2007, Word, Excel, and PowerPoint had menus and toolbars arranged differently. After the introduction of the Ribbon interface, the "Save," "Undo," and "Print" commands were placed consistently across all apps. This reduced confusion and improved learnability.
Visibility	Important options (Information/message) are hidden from the user.	*Instagram "Upload Failed"* → When posting a story or reel, Instagram sometimes just shows "Upload failed" with no reason. Users don't know if it's bad internet, file format, or a server issue. The missing detail makes it harder to fix the problem.	*Google Docs Offline Mode* → When the internet disconnects, a clear banner appears at the top: "You are offline. Changes will be saved locally." It tells the user what happened and what the system is doing, reducing confusion and panic.
Feedback	The system does not respond to user actions in a clear way. This includes, as well, showing *unclear, overly technical, or missing messages* to the user, which breaks the feedback rule.	*Mobile apps with laggy "Like" buttons* → On some social media apps (especially when the internet is slow), you tap the heart/like button but nothing changes right away. Users often tap it multiple times, which can lead to unliking/liking repeatedly or thinking the app is broken. This creates confusion and frustration.	*Modern smartphone apps* → When you tap "Send" in WhatsApp, the button changes immediately: one tick (✓) appears when the message leaves your phone, two ticks (✓✓) appear when delivered, and they turn blue when read. This progressive feedback loop reassures the user that their action worked and tells them what stage it's in.
Affordance & Constraints	Design does not suggest the correct usage	*Traditional USB connectors* → they only fit one way, but the design isn't visually obvious. Users often try to insert it upside down first. This causes frustration, repeated attempts, and sometimes even damage to the port.	*USB Type-C* → symmetrical design allows insertion in either direction. The physical constraint ensures it always works without trial and error. This reduces user frustration and aligns with intuitive use.

2. **Activity 1: ATM Interfaces Observe Bad vs Good UI [15mins + 10mins Disc]**

   We have these real ATM system UI screenshots in appendix (A)

   A. Classify these ATM screens between *Bad/Good*: Cluttered, Modern, Minimalist, Too many options, Overloaded, Confusing / Unclear messages...

   B. Identify which design rule is **violated or respected**.

   C. How does this affect the UX?

3. **Activity 2: Interface Investigation [10mins + 10mins Disc]**

   For this Activity you need to work in *groups of 3 or 4*.

   A. Students find *one bad UI they recently used (website, app, machine).*

   B. Take a screenshot, and share with the group to classify it under *Consistency, Visibility, Feedback, or Affordance*. **Justify you classification**.

4. **Activity 3: Pen & Paper Redesign Challenge [15mins + 10mins Disc]**

   For this Activity you need to work in *groups of 3 or 4*. From the previous activities or the previous lab, choose a *UI*, and focus on the existing violations.

   On a paper:

   A. Draw the current bad design;

   B. Sketch a redesign applying at least 3 rules;

   C. Focus on concept, not visuals, i.e. just use pen & paper.

5. **Activity 4: Critical Thinking & Reflection Questions [10mins Disc]**

   Now that you've completed the hands-on tasks:

   A. Which design rule do you think is most often broken in real interfaces? Why?

   B. How does breaking the *feedback* rule impact user trust in a system?

   C. If you had to redesign one of today's bad examples (ATM, printer, website, door handle), which design rule would you focus on first? Why?